



REYNOLDS EXPERIMENT

Experiment No:01

Jamima Marjan (001)¹ Abul Kalam (020)¹ Bristy Roy (024)¹ Mahmudul Kabir (029)¹

Mahir Alam Shovon (031)¹ Nabilah Hossain (050)¹ Shariar Kabir Seam (054)¹ Quazi Zidan Bokhtiar (058)¹

1 Department of Chemical Engineering, Bangladesh University of Engineering and Technology (BUET), Dhaka



INTRODUCTION

The objective of the Reynolds experiment is to determine the flow regime and verify the Reynolds number as a criterion for classifying fluid flow. Fluid flow behavior affects pressure drop, energy loss, and mixing, and can be classified as laminar, transitional, or turbulent. This study reproduces Osborne Reynolds' experiment to observe these regimes and relate them to the Reynolds number.

THEORY

The nature of flow is determined by the Reynolds number (Re), a dimensionless parameter representing the ratio of inertial forces to viscous forces. Mathematically Reynolds number is defined as

$$Re = \frac{DV\rho}{\mu}$$

Where, Re= Reynolds number
 ρ = Density of fluid (kg/m³)
V = Average velocity (m/s)
D = Pipe diameter (m)
 μ = Dynamic viscosity (Pa·s)

For circular pipes, the diameter D is used as the characteristic length as it directly represents the flow scale. For non-circular conduits, the hydraulic diameter (D_h) is used:

$$D_h = \frac{4A}{P}$$

Where, P = Wetted Perimeter (m)
A = Cross-Section Area (m²)

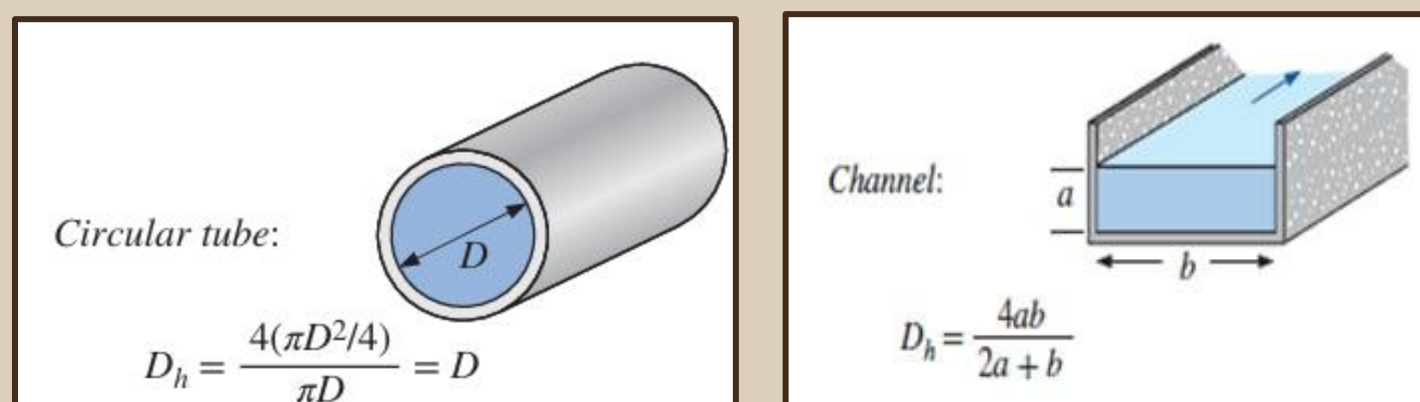


Figure 1: D_h for different structures

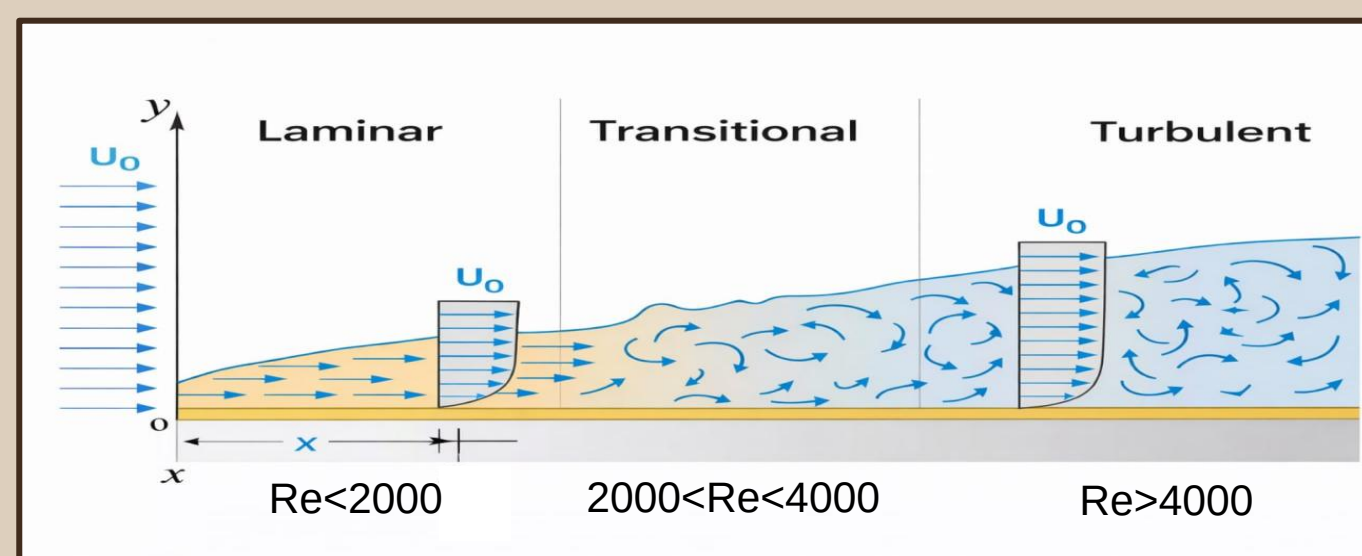


Figure 2: Different flow types and their velocity profile

In Reynolds experiment the dye is injected into flowing water to visually observe these regimes. A straight dye line indicates laminar flow, slightly dispersed line indicates transitional flow and fully dispersion indicates turbulent flow.

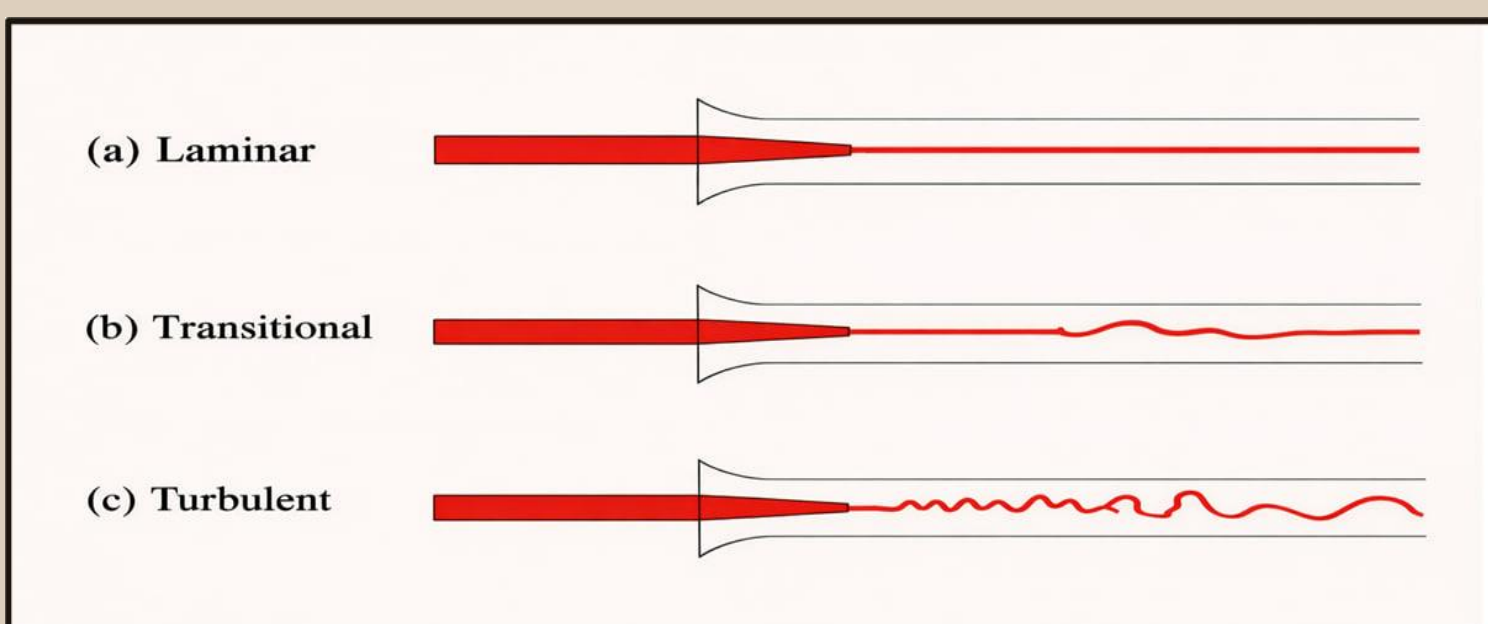


Figure 3: The behavior of injected dye in different flow regimes

ACKNOWLEDGEMENT

We express our sincere gratitude to Salman Farsi Taharat for his valuable guidance and support throughout this experiment which is conducted as a part of ChE 206 at Buet. We are also thankful to the lab instructor and our lab partners for their support in successfully conducting the experiment.

METHODS

- 1 Inject Dye into flowing water
- 2 Gradually increase flow water
- 3 Observe Dye behavior, and its pattern
- 4 Measure volume, time to measure v and Re

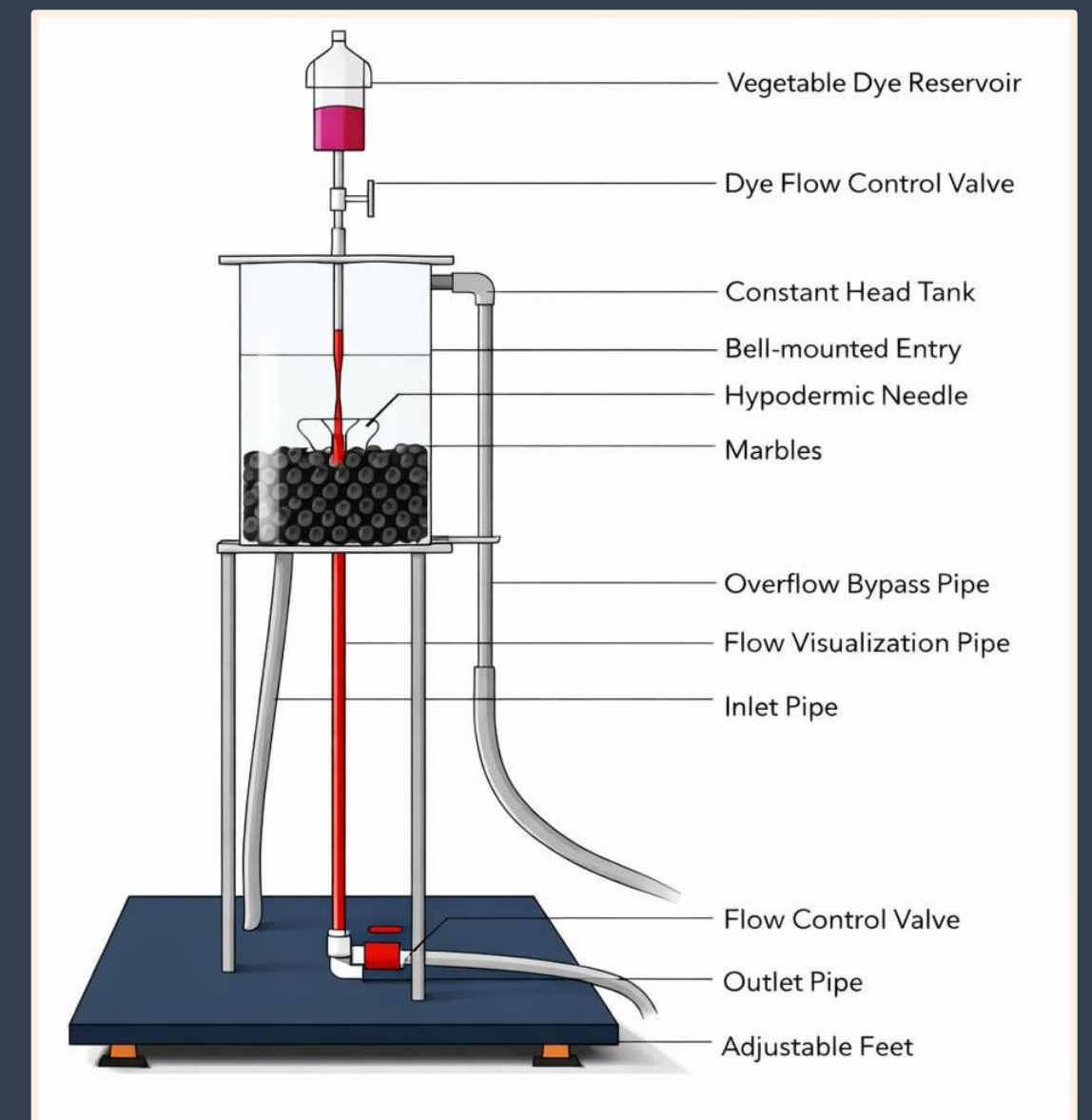


Figure 4: Schematic diagram of the experimental setup

RESULTS AND DISCUSSION

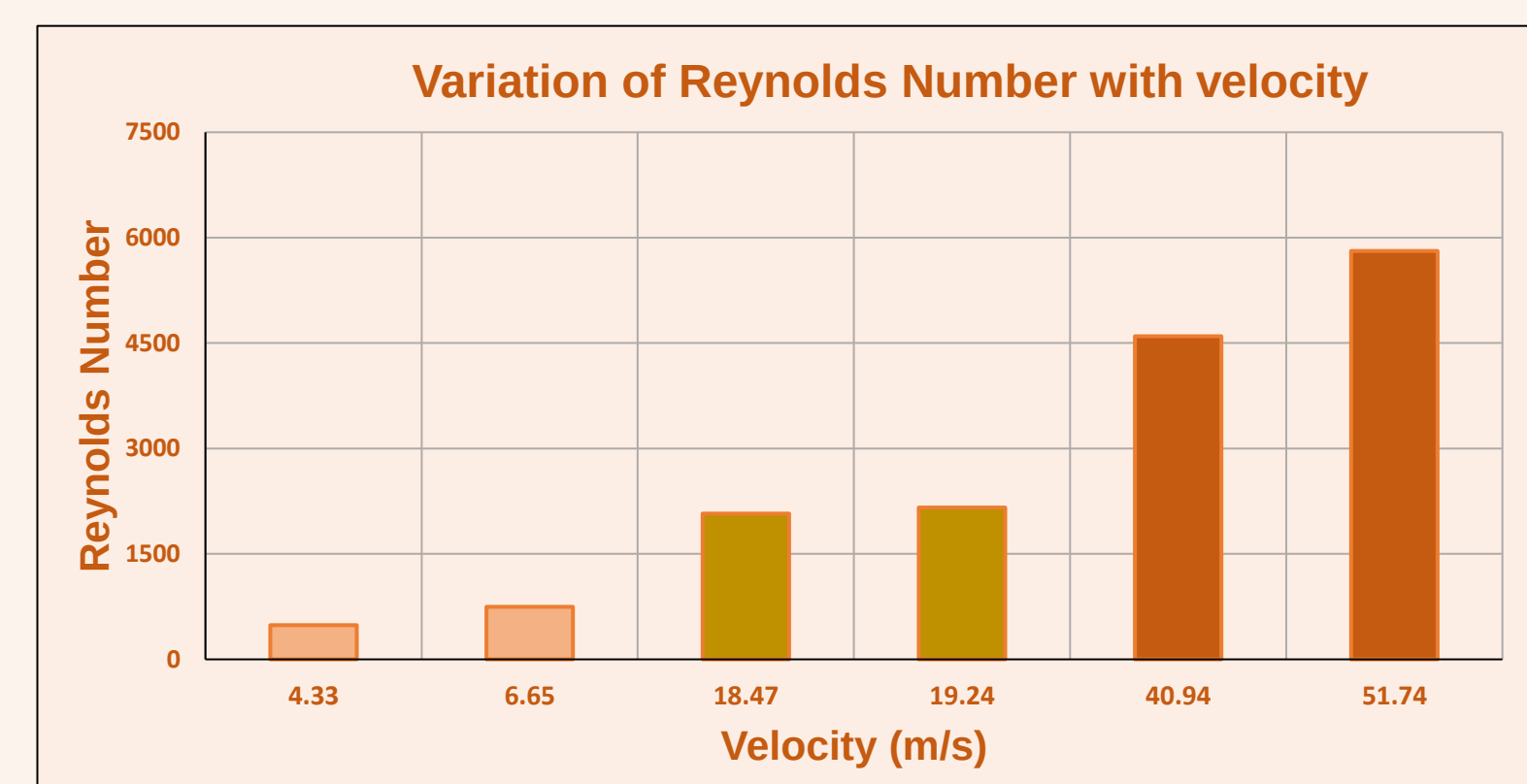
Obs. No.	Velocity $v \times 10^2$ (m/s)	Visual Dye Condition	Reynolds Number	Flow Type
1	4.33	Straight line	487	Laminar
2	6.65	Straight line	748	Laminar
3	18.47	Wavy line	2075	Transitional
4	19.24	Wavy line	2160	Transitional
5	40.94	Chaotic	4597	Turbulent
6	51.74	Chaotic	5811	Turbulent

Comments

- At low velocities, dominant viscous forces maintain the structural integrity of the dye filament confirming
- Increasing inertial forces destabilize the flow, causing the dye to oscillate and eventually disperse
- Visual transition from laminar to turbulent regimes confirms Re as a reliable indicator of flow stability.

Errors

- Changes in water temperature alter dynamic viscosity, deviating calculated Re
- Parallax error and subjective identification of flow transitions impact numerical precision
- Mechanical vibrations or un-steady dye injection can trigger premature turbulence



CONCLUSION

- The experiment successfully verified that the Reynolds number is a reliable predictor of fluid behavior, with lower values indicating laminar flow and higher values signaling turbulence.
- Observations confirmed that viscous forces maintain steady, straight flow at low velocities, while increasing inertial forces drive the transition to wavy and fully dispersed dye patterns.
- Critical Reynolds numbers deviated slightly from theoretical values due to factors like temperature variation and parallax error, the fundamental transition between flow regimes was clearly demonstrated.

REFERENCE

- [1] D. W. Green and R. H. Perry, Eds., Perry's Chemical Engineers' Handbook, 8th ed. New York, NY, USA: McGraw-Hill Education, 2018
- [2] Y. A. Ç. a. J. M. Cimbala, Fluid Mechanics: Fundamentals and Applications, New York, NY, USA: McGraw-Hill Education, 2018.
- [3] <https://share.google/ID6N7TJEPmMgXR6v>



Curious about the details ?
Scan here!